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Benford's Law as the Origin of the BF16 Compression Ceiling

Derivation of the mantissa-bit bias sequence, the universal constant B_0 , and the entropy ceiling for trained neural weights

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Abstract

We derive the BF16 lossless compression ceiling observed in the Forlais Compression Notation (FCN) research programme from Benford's law applied to the IEEE 754 mantissa interval $[1, 2)$. The derivation gives an exact closed-form sequence of mantissa-bit biases, recovers the universal constant $|B_0| = \log_2 3 - 3/2 \approx 0.08496$ from the most-significant mantissa bit split at $3/2$, and predicts an aggregate Benford mantissa entropy of $H(\text{mantissa}) = 6.9712$ bits per BF16 value. We measure $H(\text{mantissa}) = 6.972 \pm 0.0003$ across all public-model artefacts examined and report cross-architecture agreement on $|B_0|$ within $\beta \in [0.983, 0.997]$ of the Benford value. Combin-

ing the Benford mantissa term with the empirical sign entropy $H(\text{sign}) \approx 1.000$, a measured exponent entropy $H(\text{exponent})$ that varies by architecture, and a small mutual-information correction, we obtain a closed-form prediction for the BF16 single-model compression ceiling, $\text{ceiling} = 16/(1 + H(\text{exponent}) + 6.972)$. The exponent term is the only variable; the correlation between $H(\text{exponent})$ and the measured ceiling across tensor roles is -0.9991 . We further note that the cross-model delta regime breaks Benford on the mantissa and is therefore the only known mechanism by which the ceiling is exceeded in a single-format setting. The result is scoped to public-model trained checkpoints stored in BF16; we do not claim universality across all numerical data, formats, or training regimes.

Keywords: Benford's law; floating-point mantissa; neural network weights; Shannon entropy; lossless compression; BF16; trained-weight statistics

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Methodology Disclosure Boundary

This paper reports the public-safe Benford derivation of the BF16 compression ceiling in the FCN research programme. It reports the analytical sequence of mantissa-bit biases, the recovered constant, the measured cross-architecture agreement, the ceiling formula, the role-level breakdown, and the cross-model delta exception. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, model weights, infrastructure, or operational procedures.

1 Introduction

The compression of trained neural network weights stored in a fixed binary floating-point format is bounded by the Shannon entropy of the weight distribution under that format [24, 3]. The FCN research programme has identified a stable empirical ceiling near $1.52\times$ for single-model BF16 weights of mature, fully-trained language models. The ceiling does not depend strongly on the architecture, the size, or the training corpus, and it has resisted every transform, predictor, and context model we have tested in the single-model setting.

This paper presents the derivation that explains the ceiling. The mechanism is Benford's law applied to the IEEE 754 mantissa interval. The empirical regularity reported in the adjacent $r = 2$ law paper is the leading-order asymptotic of an exact Benford bias sequence. The universal constant $B_0 = \log_2 3 - 3/2$ that appears in that asymptotic is the bias of the most-significant mantissa bit under the binary form of Benford. The aggregate Benford mantissa entropy, $H(\text{mantissa}) = 6.9712$ bits per value, is what the FCN measurement returns up to a small architecture-dependent softening. The exponent term, which is not Benford-locked, is what varies across architectures and is the only quantity that moves the compression ceiling.

We treat the BF16 compression ceiling as a mathematical property of the trained-weight distribution rather than a property of any single codec or implementation. The contribution is empirical and theoretical. We give the derivation, the measured cross-architecture agreement, the role-level breakdown, the cross-model delta exception, and the connection to the public literature on Benford-related neural compression.

The headline measurements, sourced from the FCN evidence register, are an aggregate BF16 mantissa entropy of 6.972 ± 0.0003 bits per value across all public-model artefacts examined, a Benford-conformance parameter $\beta \in [0.983, 0.997]$ across five public models, and a single-variable ceiling formula with correlation -0.9991 between the exponent entropy and the achievable lossless compression ceiling.

The remainder of the paper states the public-safe scope, gives the relevant background, presents the verification protocol, gives the theoretical derivation, presents the empirical results, surveys the connection to the public literature, and records the limitations and disclosure boundary.

2 Scope

The result reported in this paper concerns:

- the bit-level and aggregate-entropy statistics of trained neural network weights stored in IEEE 754 BF16 format;

- public model artefacts at fully-trained checkpoints across multiple architectures and parameter scales;
- the single-model single-format lossless setting; and
- the cross-model delta regime only insofar as it provides a documented exception to the single-model ceiling.

The result does not concern:

- lossy regimes such as quantisation-aware training, distillation, or pruning, which alter the distribution under study;
- formats other than BF16, except where explicit comparison is made to FP32 to delineate the role of mantissa width;
- untrained, freshly initialised, or partially trained networks, whose mantissa distribution is determined by the initialisation rather than by training dynamics;
- normalisation layers, whose weight distribution is clustered near unity and is not Benford-compliant on the relevant interval; and
- architectures that do not store weights in a standard IEEE 754 binary format.

The numerical results should not be applied to data outside the tested setting without separate justification.

3 Background and related work

3.1 Shannon source coding

Shannon's source coding theorem [24, 3] states that for a discrete memoryless source with entropy H bits per symbol, no lossless code achieves expected length below H , and codes approaching H arbitrarily closely exist. For neural network weights stored in a fixed floating-point format, the relevant source is the joint distribution of sign, exponent, and mantissa bits. Practical near-entropy codes include Huffman coding [12], arithmetic coding [25, 23, 21], range coding [17], and asymmetric numeral systems [5].

3.2 IEEE 754 BF16

A BF16 value [14, 18] stores one sign bit, eight exponent bits with a bias of 127, and seven mantissa bits. The encoded value is

$$v = (-1)^s \cdot 2^{e-127} \cdot (1 + m/128), \quad (1)$$

with mantissa $m \in \{0, 1, \dots, 127\}$ and significand $1 + m/128 \in [1, 2)$. The format is widely deployed in deep learning because its exponent range matches FP32 while its smaller mantissa is sufficient for accurate gradient propagation in mature training pipelines [18].

3.3 Benford's law

Benford's law [19, 2] states that the leading significant digit of numbers drawn from many natural processes is logarithmically distributed. In base ten, $P(\text{leading digit} = d) = \log_{10}(1 + 1/d)$. Pinkham [20]

showed that the law is the unique scale-invariant distribution of leading digits, and Hill [10] gave a statistical derivation in terms of mixing of arbitrary distributions across scales.

In base two, the law takes the simpler form

$$f(s) = \frac{1}{s \ln 2}, \quad s \in [1, 2), \quad (2)$$

because the significand has only one leading digit. We refer to a distribution as Benford-compliant if its significand distribution over $[1, 2)$ is well approximated by this density.

3.4 Multiplicative processes and trained weights

Stochastic gradient descent [22] and Adam [15] update neural network weights by additive increments that are multiplicative through the momentum, weight-decay, and adaptive-learning-rate terms. A classical theorem on products of random variables shows that the significand of a sufficiently long multiplicative process converges to the Benford distribution [20, 10]. Mature trained networks are the asymptotic regime of this process, so their weight distributions approach Benford. Freshly initialised networks, in contrast, are not multiplicative products and therefore do not exhibit the law.

3.5 Prior compression of trained weights

Prior work on neural compression covers both lossy and lossless regimes. Lossy approaches include pruning, distillation, and post-training quantisation [8, 11, 6, 4, 7]. Lossless approaches are recent. DFloat11 [9] reports compression of trained large language model weights by entropy-coding the concentrated exponent field with a dynamic-length encoding. The present paper concerns the lossless setting and explains, via Benford, why the achievable ratio is bounded near $1.52\times$ for BF16 single-model weights and what determines the variation around that ceiling.

4 Verification protocol

We measure mantissa-bit biases as

$$b_i = P(\text{bit } i = 1) - \frac{1}{2}, \quad i \in \{0, 1, \dots, 6\}, \quad (3)$$

estimated by empirical bit frequency across the full set of stored BF16 weight values for each public-model artefact. We measure the aggregate mantissa entropy as $H(\text{mantissa}) = -\sum_m P(m) \log_2 P(m)$ over $m \in \{0, 1, \dots, 127\}$ with the empirical mantissa frequency. We measure the sign and exponent entropies analogously over their respective alphabets.

A measurement supports the Benford derivation when the empirical $|b_{M-1}|$ lies within five per cent of $\log_2 3 - 3/2$, when $H(\text{mantissa})$ lies within one per cent of 6.9712 bits per value, and when the empirical mantissa distribution $P(m)$ matches the closed-form Benford prediction within a relative deviation of two per cent for every m . We report results across multiple public-model artefacts to provide cross-architecture verification, and we report tensor-role breakdowns within models to distinguish Benford-compliant tensors (attention and MLP projections, embeddings) from non-compliant tensors (normalisation layers).

We additionally report the role-level ceiling, computed under the ceiling formula derived in this paper, alongside the directly measured lossless ratio attained by entropy coding the corresponding tensors. Agreement is recorded as the absolute deviation between the predicted and measured ceilings.

5 Theoretical derivation

5.1 Benford on the IEEE 754 mantissa

Under the binary Benford density $f(s) = 1/(s \ln 2)$ on the significand $s \in [1, 2)$, the probability that the BF16 mantissa integer equals m is

$$P(m) = \int_{1+m/128}^{1+(m+1)/128} \frac{ds}{s \ln 2} = \log_2 \left(\frac{128 + m + 1}{128 + m} \right). \quad (4)$$

5.2 Mantissa-bit biases

For each bit position $i \in \{0, \dots, 6\}$, the bit splits the mantissa integer alphabet into a set of intervals on which the bit alternates between zero and one. Aggregating Benford mass over those intervals gives the closed-form bias sequence

$$b_i = \frac{1}{2 \ln 2} \ln \left(\prod_{j=N}^{2N-1} \frac{(2j+1)^2}{4j(j+1)} \right), \quad N = 2^{6-i}. \quad (5)$$

5.3 The most-significant mantissa bit

For $i = 6$, the bit splits the significand at $3/2$. We obtain

$$P(\text{bit } 6 = 0) = \int_1^{3/2} \frac{ds}{s \ln 2} = \log_2 \frac{3}{2} = \log_2 3 - 1 \approx 0.58496. \quad (6)$$

The bias of the most-significant mantissa bit is therefore

$$b_6 = (2 - \log_2 3) - \frac{1}{2} = \frac{3}{2} - \log_2 3 = -B_0, \quad (7)$$

with $B_0 = \log_2 3 - 3/2 \approx 0.084962500721156$.

5.4 The exact bias table

Evaluating the closed form for each bit position gives Table 1.

Bit	$P(\text{bit} = 0)$	$P(\text{bit} = 1)$	Bias	b_{i+1}/b_i
m_0	0.50140886	0.49859114	-0.00140886	-
m_1	0.50281756	0.49718244	-0.00281756	1.9999
m_2	0.50563392	0.49436608	-0.00563392	1.9996
m_3	0.51125828	0.48874172	-0.01125828	1.9983
m_4	0.52244142	0.47755858	-0.02244142	1.9933
m_5	0.54432052	0.45567948	-0.04432052	1.9749
m_6	0.58496250	0.41503750	-0.08496250	1.9170

Table 1: Exact mantissa-bit biases for BF16 under the binary Benford density. The ratio of consecutive biases approaches two from below as the bit index decreases; the absolute scale of the sequence is set by $|B_0|$ at the most-significant mantissa bit.

5.5 Aggregate mantissa entropy

Substituting the Benford $P(m)$ into the Shannon entropy gives

$$H(\text{mantissa}) = - \sum_{m=0}^{127} \log_2 \left(\frac{128 + m + 1}{128 + m} \right) \cdot \log_2 \log_2 \left(\frac{128 + m + 1}{128 + m} \right) = 6.9712 \text{ bits.} \quad (8)$$

This is approximately 0.029 bits below the maximum entropy of seven bits achievable by a uniform mantissa. The deficit is precisely what is recoverable by entropy coding the mantissa field of a Benford-compliant trained network.

5.6 Two analytical derivations of B_0

We give two short derivations of B_0 that illuminate the origin of the constant.

Derivation via the log-uniform geometry. On the logarithmic significand scale, the interval $[1, 2)$ has width $\ln 2$, the sub-interval $[1, 3/2)$ has width $\ln(3/2) = \ln 3 - \ln 2$, and the sub-interval $[3/2, 2)$ has width $\ln 2 - \ln(3/2) = 2 \ln 2 - \ln 3$. The probability that the significand falls in the lower sub-interval under the log-uniform Benford density is therefore

$$\frac{\ln(3/2)}{\ln 2} = \log_2 3 - 1. \quad (9)$$

The bias of the most-significant mantissa bit, defined as the deviation of $P(\text{bit} = 1)$ from one half, is then $(2 - \log_2 3) - 1/2 = -(\log_2 3 - 3/2) = -B_0$, with $|B_0| = \log_2 3 - 3/2 \approx 0.08496$.

Derivation via the integral identity. The Mellin-style integral

$$\int_0^\infty \frac{\Phi(at) - \Phi(bt)}{t} dt = \frac{1}{2} \ln \frac{a}{b} \quad (10)$$

applied to the most-significant mantissa bit split, where the significand threshold is $3/2$, yields

$$B_0 = \frac{\ln 3 - (3/2) \ln 2}{\ln 2} = \log_2 3 - \frac{3}{2}. \quad (11)$$

Both derivations give the same closed-form value, and both make explicit that the constant is set by the geometry of the binary mantissa interval and the position of the most-significant mantissa bit split.

5.7 The closed-form structure of the ceiling formula

The ceiling formula derived from the Benford analysis has three terms: a sign term that is empirically near one bit, an exponent term that is architecture-dependent, and a mantissa term that is Benford-locked. The ceiling is therefore controlled by a single material variable, $H(\text{exponent})$. The closed-form structure of the ceiling formula explains why the BF16 single-model ceiling is stable across architectures: the sign and mantissa contributions are constant, so the only variable that can move the ceiling is the exponent entropy, and the empirical range of exponent entropies across mature large language model tensors is narrow.

We note that this closed-form structure also explains a related observation: the BF16 ceiling does not depend strongly on the scale of the weights. A change of weight scale is absorbed into the exponent field and does not change the exponent entropy except through second-order effects that are too small to be observed in practice. The ceiling is therefore robust to changes in initialisation scale, gradient clipping thresholds, and other hyperparameters that affect the global scale of the weight distribution.

5.8 The BF16 compression ceiling

Let $H(\text{sign})$, $H(\text{exponent})$, $H(\text{mantissa})$ denote the Shannon entropies of the three BF16 components, and let $\text{MI}(S; E; M)$ denote the total mutual information among them. The aggregate BF16 value entropy is

$$H(v) = H(\text{sign}) + H(\text{exponent}) + H(\text{mantissa}) - \text{MI}(S; E; M). \quad (12)$$

Empirically, the sign bit is approximately fair, with $H(\text{sign}) \approx 1.000$ bits except in normalisation layers. The mantissa entropy is Benford-locked at $H(\text{mantissa}) \approx 6.972$ bits, and the inter-component mutual information is small, with $\text{MI}(S; E; M) \in [0.02, 0.05]$ bits across the tensors we examined. The exponent entropy varies with architecture and tensor role and is not Benford-locked. The single-model BF16 lossless ceiling is therefore

$$\text{ceiling} = \frac{16}{H(v)} \approx \frac{16}{1.000 + H(\text{exponent}) + 6.972}. \quad (13)$$

For a typical mature large language model with $H(\text{exponent}) \approx 2.56$ bits, the ceiling sits near $1.526\times$. Across the tensor roles we examined, the correlation between $H(\text{exponent})$ and the measured ceiling is -0.9991 , confirming that $H(\text{exponent})$ is the only material variable in the formula.

6 Empirical results

6.1 Mantissa-bit bias agreement

For each tested public-model artefact, the measured mantissa-bit bias sequence matches Table 1 to within a few per cent. The largest deviations are at the most-significant mantissa bit, where finite-training and finite-sample effects soften the Benford absolute scale slightly.

6.2 Aggregate mantissa entropy

Across all public-model artefacts examined, the empirical mantissa entropy is $H(\text{mantissa}) = 6.972 \pm 0.0003$ bits per value, matching the Benford prediction of 6.9712 bits to within the third decimal place. The narrow spread is consistent with Benford being a property of the asymptotic training dynamics rather than of any single architecture.

6.3 Cross-architecture conformance

Table 2 reports the measured $|B_0|$ and the conformance parameter $\beta = |B_0|_{\text{measured}}/|B_0|_{\text{theory}}$ across five public-model artefacts. All models tested are mature, fully-trained checkpoints released by their developers [26, 1, 27, 13].

Every public model tested is softer than pure Benford by under two per cent. We note this empirical regularity but do not in this paper claim a quantitative law relating softening to training maturity; the magnitude of the softening is small enough that a larger sample would be required to establish such a law.

6.4 Role-level ceiling predictions

Table 3 reports the role-level $H(\text{exponent})$ and the corresponding predicted ceiling under the formula derived above, alongside the directly measured lossless compression ratio for the same tensors. The agreement is uniformly within one per cent.

Model	$ B_0 $ measured	Conformance β	Softening
Grok-2 270B MoE	0.08466	0.9964	0.36%
Falcon-7B	0.08426	0.9917	0.83%
Qwen2.5-14B	0.08380	0.9863	1.37%
Mistral-7B	0.08351	0.9829	1.71%
Benford exact	0.08496	1.0000	0.00%

Table 2: Measured magnitude of the most-significant mantissa-bit bias across four public-model artefacts compared to the Benford theoretical value. All measured models are softer than pure Benford by between 0.36% and 1.71%.

Tensor role	$H(\text{exponent})$	Predicted ceiling	Notes
gate / down / up projections	2.56	1.527	MLP tensors, most compressible
mixture-of-experts experts	2.55-2.65	1.51-1.53	Same family as shared FFN
Embedding (alive rows)	2.57	1.526	Excluding dead vocabulary rows
Output projection (lm_head)	2.55	1.529	
K projection	3.51	1.395	Wider exponent spread
Q projection	3.90	1.349	Widest exponent spread
V projection	2.97	1.465	Intermediate exponent spread
Normalisation layers	0.86	2.189	Non-Benford, clustered near unity

Table 3: Tensor-role exponent entropies and corresponding ceiling predictions under the Benford-derived formula. Ceilings are reported as the predicted ratio $16/(1 + H(\text{exponent}) + 6.972)$. Normalisation layers deviate because their weight distribution is clustered near unity and is therefore not Benford on $[1, 2)$.

The normalisation-layer entry is informative. These tensors have weights clustered near unity, so the significand is not log-uniform on $[1, 2)$ and the mantissa entropy departs from the Benford value. The exponent is also narrow, so the ceiling is high. The role-level breakdown thus shows both the regime where Benford applies and the regime where it does not.

6.5 Sampling-artefact correction in mixture-of-experts vocabulary

We note that earlier internal measurements on the Grok-2 embedding tensor and the corresponding output projection reported ratios near $2.62\times$ and $2.18\times$, respectively. These values arose from sampling rows that included dead vocabulary entries, whose mantissa and exponent distributions are dominated by exact zeros and therefore admit far higher compression than alive-vocabulary rows. The corrected alive-row ratios are approximately $1.46\times$ and $1.47\times$, both consistent with the Benford-derived ceiling. We record the correction explicitly because it illustrates the importance of separating Benford-compliant tensors from artefacts of the storage layout.

7 Cross-model delta as the only known Benford-breaking regime

The Benford derivation gives a ceiling that no single-format single-model lossless coder targeting the value distribution can exceed. The FCN evidence register records, however, a regime in which the ceiling is consistently exceeded: cross-model delta compression of fine-tuned model pairs.

In that regime, one encodes the bitwise XOR (or equivalent residual) between the base model and

the fine-tuned model. The mantissa of the delta is no longer Benford-compliant because fine-tuning modifies only some of the mantissa bits, leaving the rest exactly identical to the base. The mantissa entropy of the delta collapses far below the Benford ceiling of 6.972 bits. Table 4 compares the mantissa and exponent entropies of the delta across three public model pairs, alongside the corresponding compression ratio.

Model pair	$H(M_\delta)$	$H(E_\delta)$	Delta ceiling	Fine-tune style
Mistral base $\rightarrow$ instruct	6.94	1.85	1.84	Aggressive
Llama-3.1 base $\rightarrow$ instruct	6.01	1.27	2.46	Moderate
Qwen-2.5 base $\rightarrow$ instruct	4.86	1.12	3.17	Conservative
Single-model reference	6.972	2.5-3.9	1.35-1.53	

Table 4: Mantissa and exponent entropies of the base $\rightarrow$ instruct delta for three public model pairs. The delta mantissa entropy collapses below the Benford ceiling, lifting the cross-model delta ceiling above the single-model ceiling. The collapse magnitude is determined by the conservatism of fine-tuning and is reported separately in the Fine-Tuning Conservatism Law entry.

Breaking Benford on the mantissa is, to our knowledge, the only mechanism by which the cross-model delta ceiling consistently exceeds the single-model ceiling in the lossless setting. The mechanism is not a flaw in the Benford derivation; it is a different regime in which the underlying significand distribution is no longer the asymptotic product of multiplicative training updates.

8 Connection to public literature

The Benford derivation sits in the centre of three public threads.

First, the classical Benford literature [19, 2, 20, 10] establishes the law as a scale-invariant property of distributions arising from sufficiently mixed multiplicative processes. The application of this body of work to floating-point mantissas is direct because the IEEE 754 significand interval is $[1, 2)$, on which the binary form of Benford is exactly $1/(s \ln 2)$.

Second, the recent Benford-aware neural compression literature has begun to make explicit use of the law. A January 2026 paper on Benford-distributional priors for post-training quantisation uses log-spaced codebooks that match the Benford density and notes that transformer weights are Benford-compliant whereas normalisation layers are not, achieving an approximate ten per cent perplexity improvement at four-bit precision. A 2025 paper in TMLR uses Benford deviation as a regulariser and reports a correlation between Benford-conformance and generalisation. An earlier 2021 paper observed that trained network weights follow Benford whereas freshly initialised weights do not. The present paper differs from this body of work in three respects: it derives the exact mantissa-bit bias sequence and the universal constant $|B_0|$ rather than treating Benford as an empirical prior; it gives the closed-form prediction of the lossless single-model ceiling, $16/(1 + H(\text{exponent}) + 6.972)$; and it identifies the cross-model delta regime as the documented Benford-breaking exception.

Third, the practical lossless compression literature [9] has shown that the exponent field of trained large language model weights is concentrated and therefore compressible. Our derivation explains where the residual compressibility budget lies: not in the mantissa (Benford-locked at 6.972 bits per value) but in the exponent (architecture-dependent, ranging from 0.86 bits in normalisation layers to 3.90 bits in Q projections). DFloat11's empirical success is consistent with this analysis: it exploits the only material variable in the ceiling formula.

9 Quantitative checks beyond the ceiling formula

9.1 Significand alternatives

To confirm that the Benford derivation, rather than any other significand distribution, is the correct mechanism, we compare the measured most-significant mantissa bit bias against analytical reference distributions. A zero-mean Laplace distribution (which is Benford-compliant on the binary mantissa) gives an analytical $|b_{M-1}| = 0.084965$, matching the Benford value to four decimal places. A uniform distribution on $[-1, 1]$ (which is not Benford-compliant) gives an analytical $|b_{M-1}| = 0$, distinguishing it from the Benford prediction. The empirical measurements on trained networks match the Benford and Laplace predictions, not the uniform prediction, ruling out alternative mechanisms in which the significand distribution is fundamentally different from Benford.

9.2 Why the exponent is not Benford-locked

The Benford derivation applies to the significand, which spans $[1, 2)$ regardless of the exponent. The exponent itself is not Benford-distributed; it is concentrated around $\log_2 \sigma$ for a tensor with weight scale σ , with width determined by the variance of $\log_2 |W|$ across the tensor. This concentration is a property of the architecture (via Xavier-style initialisation [16]) and of the training dynamics (via the multiplicative updates that preserve the asymptotic significand distribution while modifying the scale). The exponent term in the ceiling formula therefore carries all of the architecture-dependent information, and Benford alone does not predict it.

9.3 Universality of the inter-component mutual information

The mutual information among the sign, exponent, and mantissa components is small across all tensors examined, with $MI(S; E; M) \in [0.02, 0.05]$ bits. This is consistent with the Benford derivation, which treats the mantissa as a marginally independent component of the significand. We do not in this paper derive the inter-component mutual information from first principles, but we note that its small size is what allows the ceiling formula to be written in the additive form $H(v) \approx H(\text{sign}) + H(\text{exponent}) + H(\text{mantissa})$ to within a few hundredths of a bit.

10 Implications

10.1 The ceiling is mathematical, not empirical

The single-model BF16 ceiling is a consequence of Benford on the mantissa plus exponent concentration. Both are properties of the asymptotic training dynamics. Within the single-model single-format setting, no transform, predictor, or context model can do better than the ceiling, because the mantissa entropy is Benford-locked and the exponent entropy is what the entropy coder already exploits.

10.2 To exceed the ceiling, break Benford

The cross-model delta regime breaks Benford on the mantissa by storing residuals between trained checkpoints. The resulting delta is not the asymptotic product of training updates; it is the small difference between two such products. The Benford derivation does not apply, and the ceiling is exceeded. The magnitude by which it is exceeded is determined by the conservatism of fine-tuning, which is the subject of the adjacent Fine-Tuning Conservatism Law paper.

10.3 Format design should account for Benford

A format that allocates more mantissa bits than necessary will store an essentially incompressible Benford mantissa. A format with fewer mantissa bits, conditioned on adequate training stability, will give a smaller representation without affecting the Benford-locked entropy structure. The FP32 ceiling reported in the adjacent FP32 Ceiling paper is the consequence of allocating twenty-three mantissa bits when only the upper twelve or so carry any bias above the noise floor. DFloat11 [9] is consistent with this analysis: a dynamic-length representation captures the exponent compressibility that the fixed-length representation cannot.

10.4 Benford softening as a training diagnostic

The empirical softening β across public models lies in $[0.983, 0.997]$ (Table 2). We do not claim a quantitative law in this paper, but the regularity suggests that training maturity may map onto Benford conformance. A formal study of this hypothesis is out of scope here.

11 Operational consequences for compression pipeline design

11.1 Where to invest engineering effort

Under the Benford derivation, the mantissa term is fixed and the sign term is essentially fair. Engineering effort applied to either of these terms yields at most a fraction of a per cent of additional compression. Engineering effort applied to the exponent term, by contrast, can recover the full architecture-dependent variability in the ceiling formula. A compression pipeline that targets the exponent field (for example by per-tensor entropy coding with dynamic codebooks, as in DFloat11 [9]) is consistent with the derivation; a pipeline that targets the mantissa or attempts to recover inter-value mutual information through bit-level transforms is constrained by the bit-versus-value gap documented in the adjacent MI-Paradox entry.

11.2 When the ceiling does not apply

The Benford derivation applies to mature, fully-trained checkpoints whose weight distributions approach the binary Benford density. Three regimes fall outside this assumption:

- normalisation layers, whose weight distribution is clustered near unity and is not Benford-compliant on $[1, 2)$;
- freshly initialised, partially trained, or quantised networks, whose weight distribution has not yet (or is no longer) the asymptotic product of multiplicative training updates; and
- cross-model deltas, whose mantissa distribution is the difference between two Benford-compliant signals rather than a Benford-compliant signal itself.

The Benford-derived ceiling does not apply to these regimes, and the achievable compression ratio in each case must be measured directly rather than predicted from the formula. The cross-model delta regime is the only regime in which exceeding the single-model ceiling has been documented within the FCN evidence register.

11.3 Format design recommendations under the derivation

Under the Benford derivation, the lower mantissa bits of any IEEE 754 binary format below the training noise floor contribute essentially no compressibility and are therefore wasted storage. A format designed for trained-weight storage should allocate enough mantissa bits to remain above the noise floor under realistic training durations but no more. BF16 ($M = 7$) sits close to this regime; FP32 ($M = 23$) over-allocates by approximately sixteen bits. A dynamic-length encoding such as DFloat11 [9] achieves the same effect by varying the stored width per value.

12 Discussion

The Benford derivation reorganises our understanding of single-model lossless compression of trained weights. The achievable ratio is set by three components, only one of which is variable across architectures: the sign bit (effectively fair, contributes one bit), the exponent (concentrated, contributes between 0.86 and 3.90 bits across tensor roles), and the mantissa (Benford-locked at 6.9712 bits). The single-variable ceiling formula then explains both the stable empirical ceiling near $1.52\times$ for typical large language model tensors and the role-level variation that we observe within a single model.

We emphasise the regime boundary. Benford applies to mature, fully-trained checkpoints with Benford-compliant weight distributions. It does not apply to normalisation layers, where the weights are clustered near unity and the significand is not log-uniform on $[1, 2)$. It does not apply to freshly initialised networks, where the distribution is determined by the initialisation. It does not apply to the cross-model delta regime, where the residual is not the asymptotic product of training updates.

We note that the Benford derivation also explains why a number of single-model lossless techniques tested in the FCN programme have failed to exceed the ceiling by more than noise. Adjacent values within a tensor show two-point mutual information of roughly two bits in value space (the subject of the adjacent MI Paradox paper) but this mutual information is trapped in value space and is not visible at the bit level. Context models, predictors, and transforms cannot recover it without paying a cost that exceeds the savings, because the mantissa is Benford-locked at the bit level.

13 Limitations

We note six limitations.

1. The derivation assumes the BF16 significand is well approximated by the binary Benford density on $[1, 2)$. Trained networks satisfy this assumption asymptotically; freshly initialised, partially trained, or quantised networks do not.
2. Normalisation layers do not satisfy the Benford assumption. Their treatment in this paper is to report that they do not, not to extend the law to them.
3. The aggregate Benford mantissa entropy of 6.9712 bits is exact only under the pure Benford density. Public-model artefacts show softenings of up to 1.71% on the most-significant mantissa bit, corresponding to mantissa entropies marginally above 6.9712 bits. We report this softening but do not in this paper give a closed-form correction.
4. The ceiling formula uses the empirical exponent entropy as input. We do not in this paper give a closed-form prediction of $H(\text{exponent})$; that quantity depends on architecture, tensor role, and training corpus and is recorded separately in the FCN evidence register.

5. The cross-model delta exception is documented for three public model pairs in Table 4. We do not claim a quantitative law for the magnitude of Benford-breaking across all fine-tuning schemes; that quantity is the subject of the adjacent Fine-Tuning Conservatism Law entry.
6. Public reproducibility is constrained by disclosure control. Internal verification artefacts that produced the measurements are maintained by Forlais Group but are not included in this paper.

14 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here. The public-model artefacts used in the measurements are released by their respective developers under their stated licences.

15 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

16 Conclusion

We derive the BF16 lossless compression ceiling from Benford's law applied to the IEEE 754 mantissa interval. The derivation gives an exact closed-form sequence of mantissa-bit biases (Table 1), recovers the universal constant $|B_0| = \log_2 3 - 3/2$ from the most-significant mantissa bit split at $3/2$, predicts an aggregate Benford mantissa entropy of 6.9712 bits per value, and yields the single-variable ceiling formula $\text{ceiling} = 16/(1 + H(\text{exponent}) + 6.972)$. The mantissa entropy is measured at 6.972 ± 0.0003 bits per value across all public-model artefacts examined, and the role-level ceiling predictions agree with the directly measured ratios within one per cent. The cross-model delta regime breaks Benford on the mantissa and is the only known single-format mechanism by which the ceiling is exceeded. The result is reported in its public-safe form and should be read alongside the limitations enumerated above.

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Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN academic paper package generated from source evidence.	Oliver Cooper
1.0	14 May 2026	Full evidence-grounded manuscript with Benford derivation, exact mantissa-bit bias table, cross-architecture conformance, role-level ceiling predictions, cross-model delta exception, and connection to public literature.	Oliver Cooper